Functional Specification Document of

“GASSIT”

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1. Document Control

Version History:

Version	Date	Author	Changes in version
V. 1.0	26.11.2024	Barbysheva Elena	Version for approval
V. 1.01	16.12.2024	Barbysheva Elena	Added new script “Customize User interface” in sector 4 new script “Save Progress on Object Creation” in sector 4
V. 1.02	07.02.2025	David Neugum	Revision of the entire concept for release - Addition of further Use Cases - Improvement of minor errors

Approval list

Name document section	Data	Approval	Comment
2. Introduction			
3. Project Overview			
4. Functional Requirements			
5. Non-Functional Requirements			
6. External Interface Requirements			
7. Data Requirements			

8. Workflow and Process Diagrams			
9. Reports			
10. Error Handling and Alerts			

Definition:

Term	Definition
User	An employee of a company with a specific role and rights: ● Admin; ● Main-Manager ● Rent manager; ● Office worker; ● Accountant; ● Park manager; ● Park worker. ● Customer Service
Customer	A person who has applied for a lease of premises or an existing customer who needs to extend a contract or lease additional premises.
Tenant	A customer with a valid rent agreement.
Landlord	A Laer Firmengruppe providing the customer with services for rent units
Park	An area with different types of units
Unit	Different types of premises for rent in park, including mini storage-units, garages and office spaces with kitchen facilities. (In future are more product types possible). Each product type involves different sizes, which we call Unittype.
Damage	Defect in and out of the unit or on the territory of the park
Datev	Financial accounting software
Invoice	A document indicating the money due for services rendered by Laer Firmengruppe
Debt notification	Official notice of non-payment of services
Blacklist	A list where bad customers from the past are stored so that they can no longer receive our service.

2. Introduction

Purpose of this document

The purpose of this Functional Specification Document (FSD) is to provide a detailed description of the functional and non-functional requirements for the webapplication “Gassit” for computer screens. This program is designed to streamline the process of managing and renting out units.

This system is designed to optimize the process of managing and renting out units with the ability for other companies to access the process of managing and renting out their parks of units.Automation of the process of selecting units by customers.

Development will be carried out in several stages and will include the development of:

- Web Application designed to automate the renting and managing of units;
- Mobile application for employees to optimize work processes;
- Mobile application for customers, providing the ability to independently rent and manage units;
- Scaling of the system for selling rights to use to other companies and setting up the configuration to suit the needs of the client-company..

Intended Audience:

This document serves as a reference for all project stakeholders, including developers, testers, and business analysts, ensuring that all functionalities required for the successful operation of the “Gassit” are clearly defined, understood, and can be developed, tested, and validated to meet the needs of both property managers and tenants.

Code developed so far:

A basic version of the GASSIT program has already been written. This code can and should be used as a starting point and must now be extended to the current business model. The developers use the program languages Js, vue, laravel, and java. We have a ZIP container with the previous source code including comments from the previous team.

Scope:

1. **Automate Rent Processes:** System should facilitate renting workflows, allowing to manage customer information, update property availability, process unit renting, extensions, and terminations, manage payment information and communication with customers. In the future, it should allow customers to independently select available units, reserve them, and make payments.
2. **Provide Real-Time Space Availability:** System should maintain accurate and up-to-date availability status for each unit in any location. This includes tracking whether units are vacant, under repair, reserved for specific customers, and whether they are paid or unpaid.
3. **Maintenance and Damage Tracking:** System must include features for reporting and tracking damages to rented spaces, logging maintenance requests, and assigning repair status. The process should support managing maintenance requests, assigning repair responsibilities, and tracking repair progress and associated costs, capturing customer responsibility.
4. **Utility Meter Management:** Tracking electricity usage, particularly in spaces where higher power capacities are required, calculating and invoicing consumption costs based on meter readings.
5. **Customer Status Management:** Managing customer statuses such as active, inactive, overdue, or blacklisted, as well as tracking security deposit refunds.
6. **Generate Accurate Invoices and Statements:** System will be integrated with a financial system (DATEV) to facilitate invoicing and payment processing, ensuring accurate records of rental payments, security deposits, and outstanding dues.
7. **Customer Journey Efficiency and Business Process Monitoring.** System must provide comprehensive tracking of all customer-related activities, including detailed logging of key actions with timestamps, responsible personnel, and status changes. This log should serve as a foundation for analyzing workflow efficiency, identifying bottlenecks, correcting errors, and implementing preventive measures to improve operational performance.
8. **Notifications.** System must also implement a proactive notification mechanism to inform relevant stakeholders about critical issues in real-time. These include, but are not limited to:
    a. Delays in application processing or approval stages exceeding predefined thresholds.

b. Prolonged idle status of properties, leading to revenue loss.
c. Operational errors or inconsistencies, such as incorrect status updates or unfulfilled tasks.

To support effective monitoring, System should provide customizable notification rules based on the user's role and responsibility. Alerts should be delivered through multiple channels, including email, system dashboards, ensuring timely resolution of issues.

Additionally, System should generate periodic summary reports for management and stakeholders, highlighting key metrics such as process delays, idle property units, and operational trends.

9. The reports module is designed to provide actionable insights into the financial, operational, and customer-related aspects of property rental and management. It enables stakeholders to monitor performance, track key metrics, and identify issues such as overdue payments or prolonged vacancies. Users can generate predefined or custom reports with filters by location, date, property type, or customer category. The module supports real-time and scheduled reporting, offering export options in PDF, Excel, and CSV formats. Role-based access ensures that each user views only the reports relevant to their responsibilities, maintaining security and data integrity. Availability of report types depends on the role model and is configured by the administrator or the main manager.

Out of Scope:

Not included in the description of this document and will be implemented later as part of the creation of the ecosystem “Gassit”:

1. A mobile application for employees to streamline their work processes;
2. A mobile application for customers, providing the ability to independently rent and manage spaces;
3. The implementation of functionality to enable the transfer of this desktop application rental and property management application for use by other companies.

The following areas are not covered by the process of developmemt the “Gassit”:

1. Physical security systems.
2. Door access control.
3. Real estate acquisition or property selling processes.
4. Comprehensive accounting beyond basic invoicing and payment tracking.

5. Website for customers.

3. Project Overview

Background:

Managers and administrative staff have relied on multiple, often disconnected tools to handle customer information, rent agreements, damage repair requests, and invoices.

The project was initiated to address the growing complexity and operational requirements of managing a diverse portfolio of rental properties. The customer, a property management company, oversees parks with a variety of rental properties. Each type of property serves a different tenant demographic, from individuals requiring personal storage to businesses seeking properties with specific power requirements.

Historically, managers and administrative staff have relied on multiple, often disconnected tools to handle customer information, rent agreements, maintenance requests, and billing. These manual processes have led to inefficiencies, high administrative workloads, and delays in responding to customer needs.

Objectives:

The primary objectives of the Gassit ecosystem focus on optimizing the management and rental processes, improving operational efficiency, and creating new revenue opportunities. The key objectives include:

1. **Centralized Process Control**: Implement an integrated system that consolidates all property management operations, including rent management, customer communication, maintenance tracking, and billing, into a single application. This will ensure seamless data flow, reduce the need for multiple disconnected tools, and improve overall control over business processes.
2. **Reduction in Employee Time Spent on Operations**: Automate routine tasks such as generating invoices, updating unit availability, and handling maintenance requests,

thereby reducing the time employees spend on manual operational management. This will enable staff to focus on more strategic tasks and improve productivity.

3. **Faster Turnaround for Renting Units**: Increase the speed and efficiency of the process from customer inquiry to rent agreement by implementing automated workflows and real-time data synchronization. This will minimize delays in updating unit statuses and allow for quicker occupancy by new customers.
4. **Additional Revenue Stream**: Develop and market a scalable version of the Gassit system for external companies, creating an opportunity to license the application and generate additional income. This includes customizable features that allow other property management businesses to tailor System to their unique requirements.
5. **Improved Customer Experience**: Provide a user-friendly mobile application for tenants that allows them to search for available spaces, make reservations, sign digital lease agreements, and manage their accounts independently. This self-service functionality will reduce the need for direct staff involvement and enhance tenant satisfaction.
6. **Enhanced Data Accuracy and Reporting**: Ensure real-time updates across all components of System, improving data accuracy for reporting and decision-making. This includes detailed reports on occupancy rates, rental income, maintenance costs, and utility usage, helping property managers make informed strategic decisions.
7. **Scalability and Adaptability**: Design the application to be scalable to support the expansion of the property portfolio and adaptable for future feature enhancements without significant rework. This will ensure long-term value and alignment with business growth.

The creation of the Gassit ecosystem includes the development of the following software products:

1. A web application designed to automate the rental and management of property spaces;
2. in future: A mobile application for employees to streamline their work processes;
3. in future: A mobile application for customers, providing the ability to independently rent and manage spaces;
4. The implementation of functionality to enable the transfer of this desktop application rental and property management application for use by other companies.

The first stage involves the implementation of A desktop application designed to automate the rental and management of property spaces.

4. System components

The system must run in a **cloud infrastructure (SaaS, PaaS, IaaS)**

The system is designed to automate and simplify property rental and management processes. It consists of the following key components:

- **Customer Portal (Tenant Dashboard):**
  A user-friendly interface for tenants to manage their documents, rentals, payments, and requests. The customer portal allows tenants to:
  - View a list of rented units:
  - Details for each object: name, type (mini-box, garage, office), address, area.
  - Detailed information for each object:
  - Start and end date of rent.
  - Rent price and additional expenses (e.g. for electricity).
  - Rent and payment history.
  - Current bills with the amount, payment date and status (paid/not paid).
  - Update lease terms:
  - Extend the lease term.
  - Add a new room to the already rented ones.
  - Combine several rooms (if applicable).
  - Receive notifications about important updates (e.g., payment deadlines, maintenance schedules).
  - Generate and manage access codes for properties via integration with the LLM system.
  - View events in the calendar.
  - Create new requests:
  - Request for rent of a new unit.
  - Request for repair of the rented object.
  - Request for damage.
  - Request for change of rent terms (e.g. increase of electrical power in the garage).
  - Request for termination.
  - Create a request for a meeting with the manager in the calendar.
  - View and download documents:
  - Upload documents.
  - Sign documents with an electronic digital signature.
  - Chatbot for initial communication with the customer with AI, them chat with worker;
  - Displays reports on expenses for electricity, water, payment

- **User Workspace (Internal Interface):**
  A comprehensive toolset for property managers, administrators, and other employees to manage the system's backend operations. The internal interface includes:
    - Managing tenant profiles, statuses, and requests.
    - Overseeing property statuses (e.g., rented, available, under repair).
    - Generating and managing invoices and payments.
    - Monitoring damage reports and coordinating repairs.
    - Accessing analytics and operational reports.
    - Sending, receiving and sorting letters from the mail client through the system.
    - Overview interactive plan of each park with a detailed park scheme, which allows you to:
      - see the unit status;
      - view brief information about the unit;
      - go to the page with full information about the unit or the tenant of this unit
    - Two-level Data Management: Master Data Reference and Data Reference, like:
      - Master Data Reference with Unit types and Rental Costs and deposits;
      - Master Data Reference with Damage
      - Master Data Reference Electricity and water types
      - Master Data Reference Type of tasks
      - Master Data Reference Landlords
    - Support role model:
      - Flexibility – the ability to create, edit, and delete roles without developer intervention.
      - Access Hierarchy – predefined roles with the possibility of manual customization.
      - Granular Permissions – detailed access control to specific functions and objects.
      - Independent Configuration – assigning permissions regardless of roles.
    - Making changes to all objects in the system

- **External Integrations:**
  The system is integrated with external services to enhance functionality and improve automation. Key integrations include:
    - **DATEV Financial System** for payment processing and invoicing.
    - **Credit Bureau API** for tenant creditworthiness checks.
    - **Excel** for data migration and custom reporting (input (migration all data when program starts) and output)
    - **OBDC**
    - **Property Access System (LLM):** Generates access codes for tenants and tracks usage.

- **Website** for tenant application submission and data synchronization;
- **Electronic signature module**;
- **CRM** (in future) Exchange of information on customers, sending information on calls to the system.
- **Mollie** for first payment. Acquiring system

Context Diagram:

https://drive.google.com/file/d/1lfDu8_dQGxPUAzGaj2epvDTu61YQZ6ne/view?usp=sharing

5. Functional Requirements

Overview of Functional Requirements:

The “Gassit” will include a set of core functional requirements to support the key processes involved in managing, renting, and maintaining Units. The primary functional areas include:

- **Customer and Rent Management**
  - Enable property managers to create, update, and view customer profiles, including personal information, contact details, and payment history.
  - Lock management system. Block and grant access from Gassit (integration ith LL.
  - Facilitate rent agreement management, allowing managers to create new leases, renew, or terminate existing leases, with support for multiple units per customer.
  - Allow customers to create requests for additional rental space, termination, and communicate with the customer through the customerportal, including sending notifications to the system from the customer portal and vice versa.
- **Unit Availability and Reservation**
  - Provide real-time updates on unit status (e.g., available, reserved, under repair, leased) and allow customers to view available spaces and make reservations.
  - Implement a reservation workflow where units can be temporarily reserved and held for specific customers for a limited period before the lease is finalized.
- **Billing and Invoicing**

    - Automate invoicing for monthly rent, electricity, and any additional fees, integrating with external financial systems (DATEV) for seamless payment tracking.
    - Allow managers to create and view invoices, track debts and send notifications to customers about debts, and track customer payment history within System.
- **Maintenance and Repair Management**
    - Enable tenants and property managers to report damages or maintenance issues, with workflows to categorize and assign repair tasks.
    - Track the repair status (e.g., new, in progress, resolved) and associated costs, with options for tenant accountability (i.e., identifying and billing responsible parties).
- **Electricity and Water Meter Tracking and Billing**
    - Support electricity and water tracking for type units, where necessary, including recording and reading electricity meters.
    - Generate monthly invoices for electricity costs, consumption-based billing.
- **Notifications and Alerts**
    - Send automated alerts and reminders to customers for upcoming payment due dates, lease renewals, and reservation expirations.
    - Notify employees, in accordance with the role model, of significant events such as changes in asset availability, overdue payments or unresolved maintenance requests (damage and defects), new rental requests, sending informational materials to a potential new customer, scheduling a handover meeting.
- **Security and Access Management**
    - Implement access controls, ensuring only authorized users can view or modify sensitive data.
    - Provide a secure login for tenants and property managers, with role-based access levels.
- **Integrations.**
    - Lock management system. To grant and restrict access to the unit to the customer.
    - Customers app. To be able to open the rented unit with its help.
    - Website. To be able to receive requests from customers for renting units.
    - Mail client. To be able to control correspondence and promptly notify interested parties. Perform mass mailing.

- Datev. To send invoices for payment and receive data on actual payments.
- Creditworthiness verification system. To verify the credit status and history of a potential customer

  ODBC (Open Database Connectivity) Driver

Each of these functional requirements will be developed to create an efficient, user-friendly platform for managing the diverse needs of both property managers and tenants, thereby reducing operational overhead and enhancing service quality.

System Features and Requirements

Use cases

1. Creating user in System

Precondition

The user is assigned the Administrator role.

Main script

A user with the Administrator role initiates the creation of a new user.

System:

- • suggest you to fill in the new user's details:
  - appeal to the person;
  - Last name;
  - First name;
  - Position;
  - email;
- suggest to indicate available actions In accordance with the role model for users in the System.

User with the "Administrator" role:

- enters the necessary data for the new user and the actions available to him;
- initiates saving a new user in the System.

System:

- Makes sure that the user has not been previously created in System (by checking combination of first name, last name and email);
- Saves user data;
- Generates a unique user id;

- Sends an email with a login and temporary password.

Additional script

System detected a match between the user's first and last name.

System:

- informs a user with the "Administrator" role about the presence of a user in System;
- offers to continue with the found user or create a new one;
- continues the main script with saving user data.

2. Creation of a new park

Precondition

The user is authorized in System.

Main script

The user initiates the creation of a new park profile.

System prompts us to fill in the necessary information about the new park.

User:

- Enters data about the park;
- Financial data;
- Data about the park owner;
- Creates local directories from master directories;
- Initiates saving data.

System:

- Makes sure that all required data is filled in;
- Makes sure that the data is filled in correctly;
- Adds an interactive map of the park
- Creates a new park.

Additional script

a) User does not have enough fields or have extra fields to place information about the park

User adds or deletes fields.

b) System has detected that not all required data is complete and/or incorrect.

System informs the user about the need to supplement and/or correct the data.

The user makes changes.

3. User Authentication and Authorization

Precondition

The user is registered in System. System sends an email with a login and temporary password.

Main script

The user opens the main page.

System prompts the user to enter their username and password.

The user enters a login and password.

System:

- Makes sure that the login and password are correct;
- Prompts the user to set a new password;
- Provides available features to the user.
- Opens the main page with key information relevant to a given user (DASHBOARD).

The user:

- Sets a new password
- Begins to perform his tasks in System.

Additional script

a) The user is not registered in the System or has entered an incorrect login and/or password 5 times.

System:

- Informs the user that the login and password were entered incorrectly;
- Suggests contacting the administrator to reset your password or register in System.
- Send notification to administrator for checking user.

The user contacts administrators to obtain login and password.

Administrator, if he does not find any suspicious actions, sends an email to the customerwith a login and temporary password

b) Administrator has detected suspicious actions on the part of the user.

Administrator blocks access to the system

System continues the main script by checking that all required data is complete and correct.

3.1. Creating a profile for a new units in the park

Precondition

The user is authorized in the System. A park profile has been created in the System.

Main script

The user initiates the creation of a new units profile.

System prompts you to specify information about the units (parameters, financial information).

User:

- enters data;
- initiates data saving.

System:

- makes sure that all required data is filled in;
- makes sure that the data is filled out correctly and corresponds to the parameters of the park;
- saves new units.

4. Making changes to a park or units profile

Precondition

A unit or park profile has been created in System.

Editing is available to the user.

Main script

User initiates opening the profile in edit mode.

System prompts the user to make changes.

User:

- Makes changes all data in park or unit;;
- Initiates data saving.

System:

- Ensures that all data is completed and correct;
- Saves the changes made.

Additional script

a) User changed his mind about making changes.

System does not save changes, leaving the original data.

b) System has detected that not all data is complete and correct.

System informs the user.

The user runs the main script or Additional script 1.

5. Processing the customer's application for renting a unit

Precondition

Customers need to rent one or more units.

Main script

Customer fills out a request for unit rental via the company's website.

System:

- Makes sure that the customer's data is not on the blacklist;
- Creates an application for renting units based on the requirements received through the website. (script 5.1);
- Generates a list of suitable units with offer, that includes:
    - Free units to suit customer requirements;
    - Payments (deposit, monthly rent);
    - Types of electricity and price
    - Water and price;
- Sends an email to the customer with an offer and links to confirm or reject the application;
- Starts the application processing timer (script 5.2).

Customer:

- Follows the link from the letter to confirm the application;
- Selects a unit and parameters from the offer;
- Fills in additional personal data;
- Uploads copies of documents required for renting a unit;
- Confirmation for renting the selected unit.

System:

- Saves customer data and documents;
- Sends notifications to all rent managers.

Rent manager:

- Takes the application into work (script 5.3);

- Checks the correctness of documents (script 5.4);
- Checks the customer's credit history using the system (script 5.5).

System:

- Receives a positive result of the credit history (script 5.6);
- Check and reserve units for customer in System (script 5.7);
- Create customer profile with status "Inactive";
- Generate contract with status "Draft";
- Offers the manager to complete the application and open a contract.

Rent manager:

- Check the contract
- Correct the contract if necessary;

System;

- Save changes;
- Send email to the customer with:
    - Contract,
    - Payment request;
    - Information about registration on the customer portal;
- Close the application.

Additional script

a) System did not create the request. The customer submitted a request through another communication channel.

Rent manager initiates the creation of a request in the System.

System prompts the Rent manager to fill out an application.

Rent Manager:

- If necessary, contact the customer to clarify order details;
- Enters the necessary data into the application.

System:

- Creates an application for renting units based on the requirements received through the website. (script 5.1);
- Offers the rent manager to assign the application to himself or add to the general list of applications

- Continues the main script from step “Generates a list of suitable units with payments and types of electricity and water - offer”.

b) System received a request from the CRM instead of the website

System:

- Creates an application for renting units based on the requirements received through the CRM;
- Continues the main scenario from step: Generates a list of suitable units with offer.

c) System has detected that the customer*'s data is on the blacklist*

System:

- has found a match between the customer's data and the blacklist by
  - Last name;
  - First name;
  - Address (the full list of rules will be specified during implementation);
- Adds data to the blacklist, if they new;
- Blocks the creation of the application;
- Records in the action log.

d) Customer rejected offer.

Customer:

- Follows the link to refuse;
- Selects or describes the reason for refusal;
- Confirms the refusal.

System:

- Saves the customer’s answer;
- Sends notifications to the rent manager.

Rent Manager:

- Contacts the customer to change the customer's decision;
- Suggests returning to the letter and confirming the selected unit;
- If necessary, correct the offer and resends the letter with the offer and links.

Customer continues the main scenario from the application confirmation step.

i) Customer rejected offer again.

Rent Manager:

- Indicates the final reason;

- Completes the application.

System:

- Saves the changes;
- Finished the application processing timer.

e) Customer refused to formalize the contract at any stage before signing the documents.

Rent manager:

- Indicates the reason;
- Closes the application in the System.

f) System has recorded that the waiting time for the assignment of the application has expired.

System:

- Resends the notification to the rent managers;
- Sends the notification to the main manager.
- Repeats the process in accordance with the specified rules

g) The customer needs several units

Rent manager:

- Adds additional units to the application;
- Supplements the application regarding the type of electricity, the need to combine units.
- Continues the main script with units reservation

h) The customer refused to formalize the contract at any stage before signing the documents.

Rent manager closes the application from the System.

5.1. Creating an application for renting units

Precondition

The customer fills out a request for unit rental on the company's website

Main script

System:

- Makes sure the customer is not blacklisted;

- Makes sure that the customer has not been a customer before;
- Checks and confirms the availability of the required free units in all parks of the city specified by the customer;
- Creates a new application with the customer's requirements received from the site and current data;
- Sets the status to "New";
- Sends a mail to the customer about the possibility of renting;

Additional script

a) *System has determined that the* customer *is blacklisted.*

System:

- Rejects the application;
- Sends the customer a letter about the impossibility of renting the unit.
- Stop the main script.

b) *System has detected that the customer has previously rented the units or is a current tenant.*

System:

- Sets the status to “New”;
- Sets a sign that the customer exists in System;
- Ensures that the customer has a good rental history with the company;
- Sets a sign in the customer’s profile that the units are needed for rent;
- Creates an entry in the “Chronology of Events” table in the customer profile;
- Indicates in the application the previous results of document verification;
- Send notifications to the rent managers.

Rent manager:

- Sends a proposal to the customer;
- Continues the Additional script 1.

c) *System has determined that there are no free units in the park.*

System:

- Checks for available units in nearby parks;
- Indicates that there are no free units in the selected park;
- Continues the main scenario from step “Sets the status to "New"”.

5.2. Starts the application processing timer

Precondition

System has sent notifications to rent managers

Main script

System:

- Records and saves the time spent on each stage of application processing for responsible manager, waiting for the customer's response, excludes non-working hours, weekends, holidays;
- Sends a notification to the responsible manager and the main manager about the time overrun;
- Repeats sending notifications until the conditions for changing the application status are met.

Additional scripts

a) *Responsible manager was changed*

System takes into account the total time of application processing, but keeps separate records for each responsible rent manager.

5.3. Assigning a request to a rent manager

Precondition

System has generated a new request. Sends notifications to rent managers.

Main script

Rent Manager:

- Goes to a new request from the main page (dashboard) or menu;
- Initiates transfer of the application to himself.

System:

- Ensures that the application has not been taken by another manager.
- Changes the status to “In Process”;
- Fixes the time and updates the application processing timer.

Additional scripts

b) *System has detected that the application has already been processed.*

System:

- Informs the Rent manager that the application is already in progress;
- Offers to assign the application to himself.

c) *Rent manager must reassign the selected application to another rent manager*

Rent Manager reassigned the application to another rent manager.

System:

- Change responsible manager;
- Send notification to the responsible manager;
- Change responsible manager the application processing timer.

Additional scripts

a) *Rent Manager can't reassigned the application*

Main manager reassigned the application to another rent manager.

System:

- Change responsible manager;
- Send notification to the responsible manager.

5.4. Making changes to the offer

Precondition

System creates an offer. Outside System, contacts the customer and clarifies details of the desired units and customer data:

- Electricity tariff;
- Fresh water usage for offices and work spaces;
- Rental start date;
- Lease end date;
- Access start date;
- Access end date;
- units access ID;
- Number of keys.

Main script

Rent Manager, if nessasary:

- Correcting the list of units in the offer;
- Making changes to
  - payments;
  - types of electricity;
  - rent;

System:

- Saves the results;
- Updates data in the units profile;
- Updates data in a document package.

5.5. Checks the correctness of documents

Precondition

System sends notification.

Main script

Rent Manager initiates opening or loading a document.

System is opening or loading a document.

Rent manager:

- Makes sure the document is correct;
- Confirms the correctness of the document in the system;
- Repeats actions for all documents from the list.

Additional scripts

a) Rent manager discovered that not all documents are correct

Rent manager marks incorrect documents.

System sends a letter to the customer with a link about the need to attach correct documents.

5.6. Check the customer's credit history

Precondition

Received confirmation of unit reservation and documents from the customer.

Main script

Rent manager initiates a credit history check.

System:

- Receives the result of the check using the API of the credit bureau;
- Saves the full response from the credit bureau received via the API;
- Ensures that the customer's rating is 0-1 in the credit bureau response
- Displays the digital result on the interface.

Additional scripts

a) System received a rating of 2-3 in the response from the credit bureau

System sends a notification to the general manager about the need to confirm the possibility of concluding an agreement with the customer

Main manager:

- Opens application;
- Confirmed the possibility of concluding an agreement with the customer;

System:

- Saves and displays the confirmation on interface in application;
- Sends the impossibility of renting a unit.

b) *System received a rating of 4-5 in the response from the credit bureau or main manager didn't confirm the possibility of concluding an agreement with the* customer

System:

- Saves and displays the result on interface in application;
- Rejected the application;
- Send email about the impossibility of renting a unit;
- Sends the impossibility of renting a unit.

5.7. Reserving a units for a customer

Precondition

System receives a positive result of the credit history check.

Main script

System:

- Ensures that the units is available for reservation;
- Ensures that all required details are completed and correct;
- Saves changes to the application;
- Adds the unit reserved status;
- Adds information about the reserve customer data to the unit profile.

Additional scripts

a) *System has detected that the unit is already reserved or is not available for reservation.*

System informs the rent manager.

Rent manager contacts the customer and agrees one of decision:

- Changing the unit;
- Rejecting the application

- Waiting for a new unit.

b) System could not verify the completeness and correctness of the data.

System informs the rent manager.

Rent manager continues the correctness of the data in the application.

6. Concluding a contract with a new customer

Precondition

The customer received an email with a contract and a link to register a customer profile in System.

Main script

Customer registers in customer portal at the system (Script 6.1).

System:

- Checks that the customer does not have an account;
- Creates a personal account for customer;
- Displays in the personal account:
    - Selected unit;
    - Necessary documents for renting the unit;
    - Free dates for the rental manager for meeting on transfer of unit for rent.

Customer:

- Downloads the contract and request for payment from personal account;
- Signs documents and uploads in personal account (;
- Selects the date for receiving the unit in the calendar.

System:

- Reserved the date in the manager's calendar;
- Saves document in customer profile;
- Sends notification to the responsible rent manager;
- Starts the concluding processing timer.

Rent manager:

- Confirms the correctness of the documents in the system;
- Confirms the date of the meeting with the customer.

System:

- Sends confirmation to the customer;
- Creates handover protocol;
- Sends payment information to the customer (Script 6.2)

Additional scripts

a) Customer didn't register in the customer portal.

Customer signs documents and sends them by email or letter.

Rent manager:

- Check the correctness of the documents;
- Upload documents in the system;
- Connect the customer to choose the date of meet;
- Enters the meeting date into the calendar;

System continues the main scenario 6 from step:

- Sends information about unit and payments in Datev and account.

b) System find customer account.

System prompts the customer to log into their account or restore their login details and continues the main scenario.

c) System has detected that the manager's processing time for the customer's documents has expired

System sends notification to the rent manager and main manager.

6.1. Registration in customer portal

Precondition

The customer received an email with a contract and a link to register a customer profile in System.

Main script

Customer:

- Fills out the registration form, indicating:
  - Full name.
  - Email address.

    - Phone number.
    - Password (with the required parameters, for example, at least 8 characters long, numbers and special characters).
- Agrees to the terms of use and privacy policy by checking the appropriate box.

System:

- Checks the absence of a personal account for the customer;
- Checks the correctness of the filled data:
    - Email must be in the correct format.
    - Phone number must match the format (for example, with the country code).
    - The password must meet security requirements.
- Assigns an customer ID, if don't have;
- Sends the customer a confirmation code to the specified email and/or phone number.

Customer enters the confirmation code in the appropriate field.

System:

- Checks the correctness of the entered code.
- Creates a customer account
- Redirects them to their personal account.

Customer sees information about the lease and an offer to fill in additional data (for example, sign documents).

Additional scripts

a) The system has detected incorrectness of the entered data

System offers the customer to correct the data.
customer makes changes and continues registration.

b) System has detected that the customer *already has a personal account.*

System:

- Has detected that the customer already has a personal account.
- Informs the customer and offers to enter login data

c) System has detected that the customer *already has a customer ID.*

Does not assign a new ID.

6.2. First payments for unit

Precondition

The customer received a notification about the need to pay.

Main script

System:

- Create invoice;
- Sends notification to the customer;
- Sends invoice with information to customer i:
    - Deposit amount;
    - Monthly rent amount;
    - Amount of the first rent (number of days is determined from the rental date until the end of the current month);
- Sends Link in customer account..

Customer

- Logs into the personal account;
- Follows the payment link;
- Pays the rent amount for the remaining days of the month and the deposit with 'Mollie'.

System:

- Receives payment confirmation using the API;
- Sends a notification to the rent manager about the payment and the possibility to continue the rental process and to the accountant about the payment;
- If necessary, forwards to Datev.
- Make an entry in the customer's history;
- Sets the invoice status to paid.
- Sets the status to "Paid" for the deposit in the customer's profile.

7. Transfer of unit

Precondition

The day of the meeting with the Customer arrived.

Main script

System:

- Checks for unit payment availability;
- Sends notifications to the rent manager on meeting day including payment status.

Rent manager:

- Print handover protocol from the system
- Meets with the customer near the unit on the selected day:
    - Fills out handover protocol;
    - Scans with mobile phone and download handover protocol in the system;
    - Makes with mobile phone and download unit's foto in the system
    - Sends email to customer with:
        - Handover protocol with photo;
        - Access code for LLM with instructions.

System:

- Save data in handover protocol;
- Sets the unit's status to rent.

8. Create an invoice for monthly rent payment

Precondition

Customer needs an invoice.

Main script

Customer applies for an invoice for monthly rent upon signing the contract or later.

System:

- Monthly before receiving payment, generates an invoice until the end of the contract;
- Displays it in the customer's account with the ability to download;
- If necessary, sends the invoice to the customer by email.

Additional scripts

a) Customer did not apply for an invoice

System:

- Does not create an invoice;
- Indicates only the fact of payment in the customers account.

b) Customer applies for an invoice for a previously made payment.

In the current implementation, a user who can create invoices, creates an invoice in the system.

9. Create an invoice for another payments

Precondition

Customer needs an invoice. User has the right to create an account

Main script

User:

- Creates an invoice in the system according to the event for payment, for example, payment for damage;
- Sets the invoice status to draft.

System:

- Saves the invoice status;
- Make sure that there is no invoice in the system for this event;
- Adds the invoice to the table with payments as pending confirmation;
- Does not send information to the invoice in Datev.

User

- corrects the invoice in the draft status, if necessary;
- changes the status to ready for payment.

System

- Saves the invoice status;
- Changes the status in the table with payments to pending payment;
- Sends information to Datev for payment.

Additional scripts

a) System has detected an invoice already created for the event.

System:

- Informs the user about the invoice in the system;
- Suggests making corrections if the invoice has not been sent for payment to Datev.

i) The invoice has been sent for payment to Datev.

System informs the user about the impossibility of creating and correcting the invoice

b) The invoice was generated outside System

User:

Creates a copy of the invoice in the system;

Chooses status of the invoice;

System

- Saves the invoice;
- Adds information about the invoice and its payment status to the table with payments, to the customer and unit history;
- If the invoice has the status "Awaiting payment", then adds it to the list for sending to Datev for payment.

10. Payment of monthly rent and other payments

Precondition

System creates an invoice in the system if necessary.

Main script

Customer sent a document to Sepa with payment details upon confirmation of the lease

System:

- Save payment details.
- in accordance with the schedule, transfers data to Datev:
    - On payment of monthly rent and deposit for new tenants;
    - Monthly payments of current tenants;
    - For payment of new invoices;
    - Payment arrears.

Datev:

- Receives payment in accordance with the data from the system;
- Transmits the results of the received payment to the system.

System

- Distributes the payment results by:
    - Paid;
    - Unpaid
- For overdue payments, sets the status of debt;

- Informing the debtor about non-payment (Script 10.1)

Additional scripts

a) Datev was unable to receive payment (payment details incorrect)

Datev sends information to the system.

System sends notification to the customer.

Customer sends new payment details.

Manager or account change payment details.

System sends new payment details to Datev.

10.1. Informing the tenant about non-payment

Precondition

System loaded financial information from Datev.

System has updated payment information.

System has detected a tenant who has not paid invoices and the payment deadline has passed.

Main script

System:

- Adds a renter record to the “Debt Notifications” table;
- Generates notifications of non-payment and indicates the planned delivery dates to the customer;
- Informs the accountant and main manager.
- Sends notification by email;
- Downloads a notice of non-payment;
- Sends it by mail to the customer;
- Makes a record of the event in the “Customer Interaction History” table
- Adds a comment to the tables:
    - "Debtors";
    - "Problem customers";

Notice of nonpayment is sent to the landlord three times.

Next the script is executed 2.8.3.

Additional scripts

a) When updating payment data, System detected that the invoice had been paid.

System:

- Deletes a record from the table "Debt Notifications";
- Makes a record of the event in the “Customer Interaction History” table
- Adds a comment to the tables:
    - "Debtors";
    - "Problem customers";
- Informs the accountant and main manager.

11. Changes in payment amounts

Precondition

User agreed with the main manager on the change in payments. User have the right to adjust the rental price.

Main script

User:

- Opens a section with reference information on the park;
- Selects actions to change the monthly rent, the cost of electricity and water;
- Sets up a list of units and tenants for which the changes will be applied;
- Makes changes.

System

- Saves the changes;
- Changes payment information at units;
- Sends email notifications to tenants affected by the changes;
- Makes changes to the payment information that will be sent to Datev.

Additional scripts

a) The user needs to change the payment information for a specific unit.

User makes changes to the payment information on the page of a specific unit

System:

- Saves the changes;
- Makes changes:
    - To the payment information, which will be sent to Datev, if the unit is rented.
    - Or in documents if the unit is in the process of being rented;
- Records changes in the history of the unit.

12. Settlement of damages

Precondition

Damage occurred on the territory of the park or unit.

Main script

User:

- Found damage in the units or on the territory of the park;
- Enters data about damage into System (script 12.1).

System informs the park manager.

Park Manager:

- Finds out and indicates who is responsible for damage to System (script 12.2);
- Organizes work to eliminate damage (script 12.3);
- Determines the amount of compensation for damage (script 12.4);
- Obtains consent from the person responsible to receive compensation for damage (script 12.5);
- Transfers the invoice in the system to the status "Awaiting payment";

System informs Datev about the amount of compensation for damage.

Datev receives money out of the system and sends information about payment in the system.

System closes the damage profile.

Additional scripts

a) Could not identify the person responsible.

Park manager, in agreement with the general manager:

- Closes the account in the system;
- Informs the accountant.

System:

- Transfers information on the invoice to Datev;
- Changes the account status to "Closed"

(Clarify actions in such a situation.)

b) The customer or the responsible person refuses to accept responsibility for the damage.

Park Manager, in consultation with the General Manager:

- Sends information from the system via email to the lawyer;
- Keeps the invoice status "Awaiting confirmation"
- Writes invoice to tenant of affected unit

12.1. Entering damage information into System

Precondition

The employee discovered the damage.

Main script

The employee initiates the creation of a damage profile in the System.

System prompts you to fill in the damage data.

The employee indicates the unit's details, choose type of damage, describes the damage and uploads photographs.

System:

- Establishes responsibility for the renter of the units in which damage is discovered;
- Ensures that the required data is completed;
- Records information about damage;
- Creates a record in tables:
    - “History of the units”;
    - “History of interaction with the customer” for the customer who is the renter of the units;
    - "Damage";
    - Notifies the park manager.

Additional scripts

a) The employee indicated that the damage was located within the park.

System continues the main script without specifying the person responsible and making an entry in the “customer Interaction History” table.

b) Employee found several types of damage

User on the damage page adds several types of damage, describes and uploads a photo for each type of damage.

c) The employee interrupted filling out the damage profile.

System:

- Makes sure that the required data is filled in;

- Saves the intermediate result;
- Creates a record in the “Damage” table.

User:

- Opens a damage profile from the “Damage” table or units profile;
- Continues the main script from where it left off.

d) The employee initiates deletion of the damage profile.

System deletes the damage profile (specify damage)

e) System has detected an incomplete damage profile.

System notifies the employee who initiated the creation of the profile and the park manager

f) System has detected that not all required data has been filled in.

System informs the employee that it is necessary to fill in the data to save the damage profile.

User continues the main script with filling in the data.

12.2. Determining who is responsible for the damage

Precondition

The park manager was notified of the damage.

Main script

Park Manager:

- Externally determines who is responsible for the damage;
- In the system, on the damage page, selects the responsible tenant or specifies the park visitor;
- fill on the data of insurance details of the responsible person;
- If available, fills in the external visitor details;
- If necessary, selects the tenant to which the park visitor belongs.

System:

- Saves the changes on the park page and in the tables where the damage data is displayed;
- Sends notification to the main manager.

Additional scripts

a) The park manager was unable to determine who was responsible for the damage.

The park manager removes the designated person in charge.

System:

- Saves changes;
- Corrects entries in tables (adds/changes/removes the person responsible):
  - History of interaction with the customer;
  - History of the units;
  - "Damage".

b) The park manager has determined that another renter is responsible for the damage.

The park manager changes the person responsible for the damage profile.

System performs the steps from the Additional script a.

c) The manager of the park identified an external visitor, who is not a renter, as responsible.

Park Manager:

- Finds out the contact details of the person responsible for the damage;
- Changes the person responsible in the damage profile;
- Indicates which renter the responsible person is associated with.

System:

- Establishes an indication that the person responsible is not a renter;
- System performs the steps from the Additional script 1.

12.3. Repairing damage

Precondition

The fleet manager determined, outside System, how to repair the damage.

Main script

The park manager indicates on the damage profile:

- A sign that the repairs are being carried out by the landlord;
- A park employee who will repair the damage;
- Date of start and completion of repairs.

System sends a notification to the park employee.

Park Manager:

- With the help of a park employee, organizes work to eliminate damage;
- Accepts the results of repairs;
- Indicates on the profile that the damage has been repaired.

System:

- Indicates the Damage Cleared event in the tables:
    - History of interaction with the customer;
- History of the units;
- Deletes a record from the “Damage” table

Additional scripts

The park manager indicated that the repairs were being carried out by the customer.

The park manager indicates the start and completion date of repairs on the damage profile.

System informs the customer by email.

The customer repairs the damage.

Park Manager:

- Accepts repair of damage by the customer;
- Initiates closing of the damage profile in System.

System:

- Records the closure of the profile;
- Continues the main script from the step of specifying the “Damage repaired” event in the tables.

The park manager did not accept the repair of the damage.

The park manager indicates on the damage profile that the damage has not been repaired.

The contractor repairs the damage again.

The fleet manager continues the main script with the step of receiving the repair result.

The scenario is repeated until the damage is repaired or the contractor is changed (recorded in the history of the units).

The park manager indicated that repairs were not being carried out.

System makes an entry in the history of the unit and the tenant.

12.4. Determines the amount of compensation for damage

Precondition

Damage was created in the system.

Main script

Park manager:

- Creates an invoice in the system;
- Selects the type of damage from the directory;
- Sets the area or size of damage;
- Initiates saving the invoice.

System:

- Calculates the amount of payment for damage;
- Saves the invoice;
- Adds the payment to the "Payments" table;
- Sets the status "Awaiting confirmation".

12.5. Sending an invoice for payment

Precondition

Park manager indicated the method of elimination by the lessor or without repair.

Main script

The park manager enters the following on the damage profile:

- Units of damage measurement;
- Number of units;
- Unit cost.

System:

- Calculates the cost of compensation from reference data ;
- Forms texture.

Park Manager:

- Agrees on the amount of compensation with the main manager;
- Obtains consent from the customer for payment;
- Sends an invoice for payment or chooses to draw it from the customers bank account via direct debit.

System executes the script “Payment of monthly rent and other payments”

Additional scripts

a) The park manager did not agree on the cost of compensation.

Park manager, in consultation with the general manager, makes changes to the invoice if necessary and repeats the main script again.

b) System has detected that the responsible renter already has non-payment or damage events.

System:

- Creates a record in the “Problem customers” table;
- Sets the status “Problem customers” in the customer profile.

13. Termination of the rental agreement

Precondition

Customer wants to terminate the contract.

Main script

Customer in the personal account:

- Creates a request to terminate the rent agreement with data:
    - Date of termination;
    - Reason of termination.
- Indicates date of transfer of units in calendar.

System:

- Make sure that there are more than three months left until the contract termination date;
- Make sure that the manager's time is free in the calendar;
- Adds a meeting to the task list and the manager's calendar;
- Sets the statuses for the unit and the customer about the contract termination;
- Indicates that the Unit is available for rent.

Rent manager:

- Confirms the meeting in the calendar or task;
- Meets the customer on the appointed date to accept the unit (Script 13.1);
- Fills out the handover protocol for termination with the customer (Script 13.2);
- Close the termination;
- Sets a mark for the return of the deposit.

System:

- Sets the status of the contract “Terminated by the customer”;
- Sets a mark in the customer's profile that the deposit is awaiting return

- Sends a notification to the accountant about the need to check the debt on payment and return the deposit to the customer (Script 15).

Account returned deposit

Additional scripts

a) Customer informs about the termination via email

Rent manager:

- Makes sure that there are more than three months left until the contract termination date;
- Initiates the termination in the system;
- Indicates in the customer profile:
  - Date of termination;
  - Date of transfer of units;
  - Sign that the termination was initiated by the renter.
- Outside the system, he clarifies with the customer the date of the unit transfer;
- Indicates the date and time of the unit transfer in the calendar.

System:

- Makes sure that the manager's time is free in the calendar;
- Creates a meeting in the customer's calendar in the personal account;
- Adds a meeting to the task list and the manager's calendar;
- Sets the statuses for the unit and the customer about the contract termination;
- Indicates that the Unit is available for rent.

Rent manager:

- Confirms the meeting in the calendar;
- Meets the customer on the appointed date to accept the unit;
- Makes sure that the customer has no debt;
- Sets a mark for the return of the deposit.

b) System has detected that there are less than three months left until the termination date.

System informs the user and offers to select another date.

c) *System has detected that the time and date selected by the* customer *for transferring the unit is occupied by the manager.*

System offers options for available manager time closest to the selected date.

d) *The Rent manager discovered a debt from the customer.*

The Rent manager informs the customer about the deduction of debt from the deposit for the units.

e) *The tenant terminates the lease agreement early.*

Rent manager executes main script.

f) *The customer extends the contract for another year in a response letter.*

System/Rent manager changes the end date of the lease agreement.

d) The Rent manager did not accept the units.

Manager:

- schedules a follow-up meeting to hand over the unit;
- records the reason in the system.

13.1. Acceptance of units from the customer

Precondition

The Rent manager and the customer met to transfer the units.

Main script

Rent manager:

- Inspects the units;
- Fills out the “Hand over protocol” units acceptance form in System;
- Make photo and upload them in the system;
- Sets a sign that the unit has been accepted.

System:

- Records the results of the reception of the units;
- Changes the units status to “Free”;
- Deletes a record from the “Termination” table;
- Change a record in the table “Units”.
- Creates a lease end record in the tables:
    - “History of the units”;
    - “History of interaction with the customer.”

Additional scripts

a) The Rent manager discovered damage during an inspection of the units.

The Rent manager indicates the presence of damage on the ABNAHME units acceptance form.

System continues scenario “Settlement of damage” with the step of informing the fleet manager.

b) The customer transfers the repair of damage to the renter.

Rent manager:

- Informs the customer that the amount of damage will be deducted from the deposit or an invoice will be issued;
- Continues the main script with an indication that the units have been accepted and repairs are needed.

System:

- Records the results of the reception of the units;
- Changes the units status to “Inactive”;
- Deletes a record from the “Termination” table;
- Creates a record in the table “Inactive units”;
- Creates a lease end record in the tables:
- “History of the units”;
- “History of interaction with the customer”;
- Informs the park manager.

14. Termination of the contract at the initiative of the landlord

Precondition

The customer systematically fails to pay rent and/or damages the units.

Main script

The Rent manager sets in the customer's profile the indication of termination at the initiative of the lessor and the date of termination of the contract.

System:

- Generates three notifications for the customer;
- Sends the first notification to the customer;
- Sets the date for sending all notifications to the customer’s email;
- Duplicates the date of termination in the units profile;

- Creates a record in tables:
- “History of the units”;
- “History of interaction with the customer”;
- "Termination";
- Forms a units acceptance form;

The Rent manager, after sending three notices and the termination date.

15. Return deposit

Precondition

System sends a notification to the accountant about the need to check the debt on payment and return the deposit to the customer

Main script

System creates a task for the accountant to return the deposit;

Account:

- Checks the debt on payment to the customer;
- Outside the system, through the bank, the Accountant returns the deposit to the customer;
- Enters information about the return of the deposit into the system.

System:

- Saves information about return of the deposit in customer profile and Vendors table
- Closes the contract
- Sets the customer's status to "Inactive"
- Sends notification to the customer and rent manager.

Additional scripts

a) The accountant did not change the status of the deposit within three days.

System:

- Sends a notification to the accountant about the need to return the deposit or indicate the reason for the delay in return;
- Sets the status in the Vendors table to not fulfilled for this deposit.

Accountant:

- Opens a request for a deposit return;

- Indicates the reason for the delay;
- Indicates the expected date of return of the deposit

If the deposit is not returned on the specified date, the scenario is repeated

16. Receipt of the debt on the account

Precondition

System received information from Datev about non-payment of invoice.

Main script

System:

- Sends reminder by letter to customer and manager every week;
- Adds customer to table with debtors;
- In the table "Payments" sets the status of invoice "Debt".

customer is late with payment of invoice for more than one month.
Manager calls the customer during this time and writes a comment on the table “History of customer’.

System:

- Informs the manager, general manager;
- Sends an official notice of non-payment by email;
- Adds the status “Problem” in customer
- Creates a record in the “Problem customers” table;
- Sends a copy of the notice to the Office worker.

Office worker:

- Sends an official notice of non-payment by e-mail to the customer outside the system;
- Uploads a receipt into the system confirming that the letter was sent to the customer

System and the Office worker repeat the process three times with the official notice sent.

System:

- Sends a final notice to the manager, general manager;
- Sends a package of documents by e-mail to the Lawyer with a copy to the manager and general manager.

Manager initiates the process of terminating the contract with the customer (Script 14). Lawyer will settle the payment issue outside the system.

System:

- Receives confirmation from Datev about payment of the invoice;
- Changes the status to "Paid".
- Deletes the customer from the "Debtors" table;
- Adds the status “Problem” in Customer
- Creates a record in the “Problem customers” table;

Additional scripts

a) Customer has paid the invoice in progress.

System:

- Interrupts the debt collection process;
- Sends a notification to all users involved in the process;
- Deletes the customer from the "Debtors" table.

b) Failed to receive payment for the invoice.

Manager, in agreement with the general manager, closes the invoice.

System:

- Sets the invoice status to "Unpaid closed";
- Does not send any more information to Datev about waiting for payment for this invoice;
- Deletes the customer from the "Debtors" table;
- Saves the customer in the "Black List" table.

c) Payment for the invoice has been made after the invoice has been closed.

System:

- Sets the invoice status to "Paid";
- Saves the customer in the "Black List" table.

17. Customize User interface

Precondition

Admin must be logged in to the system and have the appropriate permissions to access the user field customization module.

Admin selects the "Customize Forms" option in the administrative panel menu.

Main script

Admin Initiates the session by accessing the customization module via the administrative panel.

System displays the customization interface, including options for adding, modifying, and deleting fields.

Admin selects the "Add Field" option to introduce a new field to a form.

System presents a dialogue with field options including type (text, dropdown, checkbox, etc.), name, default values, and validation rules.

Admin:

- Configures the new field's parameters;
- Submits the information.

System:

- Validates the provided settings;
- Adds the new field to the database;
- Updates the form in real-time to include the new field;
- Confirms to the Administrator that the field has been added;
- Make sure that the field was not previously in the database;
- Logs the changes made in the action log.

Additional scripts

a) Admin chooses to modify an existing field by selecting it from a list of current fields.

System displays the current settings of the selected field for modification.

Admin:

- Makes necessary changes to the field settings;
- Submits the update.

System:

- Processes the modifications;
- Updates the field details in the database for all related objects (e.g. document templates, letter templates, customer application);
- Visually updates the form;

- Logs the changes made in the action log.

b) Admin decides to remove a field and selects the "Delete" option for a specific field.

System asks for confirmation to ensure the deletion is intentional.

Admin confirms the deletion.

System:

- Removes the field from interface;
- Leaves unchanged the data that was filled in the field in the database
- Updates the form to reflect this change;
- Logs the changes made in the action log.

Admin reviews the updated form to ensure all changes are applied correctly.

System provides a visual confirmation that the form now reflects all recent modifications.

c) System has detected that the field was previously on the interface.

System:

- Informs the administrator that the data will be restored;
- Offers to select a solution for objects that are in the system, but there is no data:
  - Make the field optional;
  - Fill with the default value;
  - Perform a mailing to customers or create a notification in the personal account, if the data requires additional information from the customer .

Admin selects the action.

System:

- Performs the action in accordance with the administrator's choice;
- Logs the changes made in the action log.

18. Save Progress on Object Creation

Precondition

User is authenticated in the system..

Main script

User begins filling out the object creation form.

System displays the form and starts tracking data entry.

User enters available data into several fields.

System automatically saves the entered data at regular intervals to temporary storage.

User:

- Interrupts the filling process;
- closes the browser or logs out of the system;
- Returns to complete the object creation form at a later time.

System:

- Automatically identifies the customer;
- Provides the last saved state of the form with previously entered data.

User:

- Continues filling out the form from where they left off;
- Completes the form and confirms the data.

System saves the final data to permanent storage and removes the temporary data.

Additional scripts

a) System detects a connection loss.

If the system detects a loss of connection during an autosave attempt, it tries to save again and notifies the customer to check their connection.

b) User detects restored data errors

User:

- Choose to start over or edit specific fields;
- Continues to fill in the object data in accordance with the main script.

19. Control of communication with the customer

Main script

Employee makes a call to the customer or sends an email.

System records the fact of communication with the customer using the API CRM or an email customer

Employee, if necessary, opens previous letters to the customer in the system.

20. Problem customer

Precondition

Employee found that the customer violated the rules of the renter.

Main script

User:

- Discovered that the customer violated the tenant rules;

- Added the "Problem" status in the customer's profile;
- Creates a record in the “Problem customers” table;
- Added a description, if necessary;
- Initiated saving.

System:

- Saves the changes made;
- Adds a time period during which the customer will be "problem".

Customer has not violated the rules again.
System removes the customer from the problem after the time period has expired.

Additional scripts

a) Client has violated the rules again

The Employee contacts the tenant and terminates the contract. If the customer caused major problems, the employer/user can add the customer on the blacklist.

21. Assigning tasks to user and monitoring their execution

Precondition

Main manager logs into the system and navigates to the "Ticket Management" section.

Main script

Main manager:

- Selects the "New ticket" option.
- Fills out the task creation form:
    - Ticket Name (e.g., "Electrical Wiring Repair in Garage Unit 102").
    - Type work (choose from reference, e.g., cleaning, repair and so on).
    - Ticket Description (e.g., "Replace faulty wiring in the garage. Ensure proper functionality after replacement.").
    - Assigned Employee (e.g., "John Doe").
    - Linked Unit (e.g., Unit 102 in Park A).
    - Ticket Priority (e.g., High, Medium, Low).
    - Time period (e.g., "2024-02-15 18:00").
    - Upload photos or documents, if needed.
- Confirms the task by clicking the "Save and Assign" button

System:

- Sends a notification to the assigned employee about the new task.
- Marks the ticket as "New" in the task table.

User:

- Receives the task in their dashboard (can view details, descriptions, and the deadline).
- Clicks accept work.
- Upload photos or documents, if needed.
- As the task progresses, the employee updates the task status:
    - In Progress: The task has started.
    - Needs Clarification: Additional information or resources are required.
    - Completed: The task is finished.

System logs status changes and sends a notification to the manager.

Manager verifies the task completion:

- If completed correctly, they mark the task as "Closed".
- If further work is needed, they return the task to the employee (status: "Returned for Revision").

System after successful completion, the task is recorded as "Completed" with the date and time of completion.

Additional scripts

a) Employee marks the task status as "Unable to Complete".

System requires the employee to provide a reason (e.g., "insufficient resources").

Main manager reviews the task and either:

- Assigns a new executor.
- Provides the necessary resources to complete the task.

b) The task's deadline expires, but the task is not completed.

System:

- Sends a notification to the manager about the delay.
- Marks the task as "Overdue".

Main manager:

- Contacts the employee to identify the cause.
- Adjusts the deadline or reassigns the task if necessary.

c) Ticket require chief manager approval before execution

System sends notification on the date for confirmation.

Main manager is confirming or cancelling the ticket.

22. Determination of charges for consumed electricity, water

Main script

Park worker:

- Opens the park page;
- Selects the action “Make changes to meter readings”;

System prompts you to specify meter values for each unit in the park.

The park employee alternately indicates the meter values for each unit in the park.

System:

- Saves the new value and the date the value was entered;
- Calculates and indicates in a table the amount of electricity consumed per month;
- Updates/generates payment invoices for each units;
- Creates a record in tables:
    - "New invoice";
    - “History of the units”;
    - “History of interaction with the customer”, for the renter of the units.

Additional scripts

The park employee interrupted making changes.

System saves the changes made by the park employee.

The park employee decided to continue making changes.

System opens the table in the units where the employee interrupted.

The park employee continues the main script by indicating the counter value for each unit.

23. Editing a Reference Book Entry

Precondition

User (e.g., admin) have rules to change "Reference Books" section.

Main script

User:

- Selects the desired reference book from the list (e.g., "Property Types").
- Clicks the "Add New Entry" button or the "Edit" button.

System displays a form for entering data.

User:

- Completes or edits required fields (e.g., property type name, description, additional attributes).
- Optional fields (e.g., images or documents) can also be added if needed.
- Clicks "Save".

System

- validates the data (e.g., checks for unique names). If the data is valid, the system saves the entry and displays a success message.
- Adds the new entry to the list.
- Refreshes the display.

Additional scripts

a) Invalid Data.

If the input data does not meet requirements (e.g., duplicate names), the system displays an error message and prompts the user to correct it.

b) Cancel Action

User can click "Cancel" to return to the list without saving changes.

c) Conflict with Existing Data

If another entry already contains the same data (e.g., duplicate tariff names), the system notifies the user and prompts for correction.

d) Selecting an Entry to Delete:

User selects the desired entry from the list and clicks the **"Delete"** button.

System removes the ability to select this type in the reference for new tasks, but leaves it in tasks where it is already used and retains the ability to generate reports with this type

6. Non-Functional Requirements

Hosting

1. The system must run in a **cloud infrastructure (SaaS, PaaS, IaaS)**, ensuring **flexibility, scalability, and fault tolerance**.
2. Access to the system should be provided via a **web interface**, eliminating the need for local software installation.
3. User data and files must be stored in **secure cloud storage**, with support for **automatic backups**.
4. The cloud environment should support **elastic scaling**, allowing for increased computational resources based on demand.
5. The system availability level must be **at least 99%**.
6. **Data protection mechanisms** must be implemented, including **encryption, backups, and access control**.

Performance:

[Performance requirements such as response time, throughput]

1. At the first stage, System must support up to **50 active users** concurrently.
2. After implementing the customer module, System must scale to support up to **200 active users** **simultaneously**.
3. In the final version, System must handle **500 or more active users** simultaneously without significant performance degradation.
4. 80% of users will perform low-intensity tasks (e.g., browsing information, searching for units).
5. 20% of users will actively perform high-intensity operations (e.g., creating invoices, updating rental statuses, generating reports).
6. Response time for key operations (e.g., status updates, data searches, update payments) must not exceed:

    - 0.05 seconds for page loads.
    - 0.5 seconds for report generation.
7. Report generation time:
    - For small data volumes (up to 1 month) — no more than **0.5 seconds**.
    - For large data volumes (up to 5 years) — no more than **5 seconds**.
8. Full loading of user pages and interfaces:
    - On desktop devices — no more than **0.05 seconds**.
    - On mobile devices and outdated browsers, mobile — no more than **0.1 seconds**.
9. System must update statuses (e.g., "rented," "paid") in real-time when corresponding actions are performed by the user or data is received from integrations (e.g., DATEV).
10. Support for stream processing of transactions with external systems, such as the DATEV financial system or LLM for generating access codes.
11. Local caching must be used to speed up the loading of frequently accessed data (e.g., property lists, rental statuses).
12. If thresholds are exceeded (e.g., response time exceeds 2 seconds), System must notify administrators via email or integrated monitoring tools.
13. System must be tested to handle loads 50% higher than expected (e.g., 750 active users for the final version).
14. System must be tested for performance under failure conditions, such as external system delays.
15. After System launch, performance must be monitored for the first 3 months to identify bottlenecks and adjust the architecture.(? need aproove)

Usability:

[Requirements for usability, interface and accessibility]

1. The interface must support both desktop and mobile devices, ensuring responsive design for laptops, tablets, and smartphones. System must correctly display the User interface with a resolution of 2048×1080 pixels.
2. The interface must be intuitive for users with no prior technical expertise.
3. Core functions, such as creating rentals, updating property statuses, and generating reports, must be accessible within no more than 3 clicks.
4. The user interface response time for core operations, such as search or status updates, must not exceed 0.5 seconds.

5. All animations and transitions in the interface must be smooth, with a refresh rate of at least 60 frames per second.
6. System must support interface translation into multiple languages (e.g., English, German).
7. System must support regional settings (date format, currency)
8. Admins must be able to customize Users dashboards by adding or removing widgets based on their tasks.
9. System must automatically save user preferences, such as language selection, filters, or preferred reports.

Reliability:

[System reliability requirements, e.g., uptime]

1. The system must have an overall annual availability of 99%. A failure must not exceed 5 hours at a time.
2. System updates and maintenance must be performed during pre-scheduled periods with at least 48 hours' notice to users. Updates and maintenance are done during nighttime.
3. System must operate stably even during peak load periods, such as the end of the month or mass submission of applications.
4. All critical system components must have backup servers (replicas) to minimize the risk of failure. Or Automatic data backups must be performed daily, with data retention for the last 30 days. A complete backup every 30 days and record changes every day. Keep backup-copies for 3 or 6 months. Clarify memory size and cost and make a decision.
5. After a complete failure, System must be fully restored within no more than 4 hours.
6. System must handle the conflict properly, ensuring data integrity and maintaining its functionality, if two or more users perform the same actions simultaneously.

Security:

[Требования к защите данных и пользователей]

1. System must support access segregation based on roles, such as administrator, manager, tenant, and technical specialist.
2. Each role must have access only to specific system functions and data.
3. User sessions must automatically terminate after 15 minutes of inactivity.

4. Admins must have the ability to manually terminate sessions.
5. Data, including customer personal information, financial data, and property details, must be encrypted.
6. All data backups must be encrypted and protected from unauthorized access.
7. System must log all user activities, including login, logout, data changes, and critical operations (e.g., rental status updates).
8. Logs must be retained for a minimum period (e.g., 6 months) to meet audit requirements.
9. Logs must be accessible only to authorized administrators.
10. System must comply with GDPR requirements, including the ability to delete user data upon request.
11. Financial data and contracts must be processed in accordance with regional legislation (e.g., accounting rules in Germany.
12. Access to data from different locations must be isolated. Users can only view the properties they manage. And the ones they have been granted viewing and/or editing access
13. System must be resilient against various types of threats to ensure data integrity, availability, and operational security.

Compatibility

Compatibility with popular browsers, including outdated versions (e.g., IE11), must be maintained (MS Internet Explorer, Mozilla Firefox, Google Chrome, Safari, Opera).

customer Devices:

- Support for PCs with operating systems Windows 10+ and macOS 11+.
- Compatibility with tablets and smartphones running Android 7+ and iOS 11+.

Printers and Input Devices: (It is advisable to specify the equipment with which System will interact)

Compliance:

EU General Data Protection Regulation (GDPR)is very important.

7. External Interface Requirements

User Interfaces:

Main menu

Description:

The main menu will be implemented as a "Hamburger Menu" and will expand into a vertical sidebar menu.
System menu must be adapted to user roles. Each role should have configurable menu items, including access level management (view, edit).

Structure:

- Dashboard:
- Workroom:
    - New application;
    - New damage;
    - New ticket;
    - New invoice
    - Tasks
- Request:
    - Applications
    - Waiting list
    - Report (Requests)
- Customer:
    - Customer
    - Blacklist
    - Report (Customer)
- Units
    - Units
    - Damage
    - Report (Unit)
- Park
    - Location

    - Electricity / Water Usage (type reference and general price change)
    - Types
    - Plans
    - Discount
    - Insurance
- Finance:
    - Payments
    - Debtors
    - Vendors (deposit)
    - Report (Finance)
- Reports
    - Unit reports
    - Customer reports
    - Finance reports
    - Employee reports
- Mail
    - Mass Mailing
    - Schedule mails
    - Sent mail
- Settings
    - Dashboard settings
    - Document templates
    - Mail templates
- Admin
    - Users (customer)
    - Employees
    - Settings

Role model

	Admin	Main Manager	Rental Manager	Park worker	Account	Office-worker
Dashboard	x	x	x		x	x
Workroom	x	x	x	x	x	x

New application	x	x	x	x		x
New damage	x	x	x	x		x
New ticket	x	x	x			x
New invoice	x	x	x		x	x
Tasks	x	x	x	x	x	x
Request	x	x	x	x	x	x
Applications	x	x	x			x
Waiting list	x	x	x		x	x
Report (Requests)	x		x			
Customer	x	x	x	x	x	x
Customer List	x	x	x	x	x	x
Blacklist	x	x	x		x	x
Reports (Customer)	x		x		x	
Units	x	x	x	x	x	x
Units	x	x	x	x	x	x
Damage	x	x	x	x	x	x
Report (Units)	x		x		x	
Park	x	x	x		x	x
Location	x	x			x	
Electricity	x	x			x	
Types	x	x	x			x
Plans	x	x	x		x	

Discount	x	x				
Insurance	x	x				
Finance	x	x			x	
Invoice	x	x	x		x	
Payments	x	x	x		x	
Debtors	x	x	x		x	
Report (Finance)	x				x	
Reports	x	x	x		x	
Unit reports	x	x				
Customer reports	x	x				
Finance Reports	x	x				
Damage Reports	x	x	x		x	
Productivity Reports (Tasks of employees)	x	x				
Mail	x	x	x	x	x	x
Mass Mailing	x	x	x		x	x
Schedule mails	x	x	x		x	x
Sent mail	x	x	x	x	x	x
Settings	x	x				
Dashboard Settings	x	x				
Document templates	x	x				
Mail templates	x	x				

Admin	x	x				
Users (Customer)	x	x				
User (Employees)	x	x				
Settings	x	x				

Dashboard

Description

The dashboard serves as the central interface of System, providing users with quick access to key information, operational tasks, and analytical data. Its primary goal is to offer a personalized and intuitive interface for managing processes efficiently.

The dashboard should be:

- adaptive;
- customizable by the administrator;
- tailored to the user's role;
- information must be updated in real time.

The top of the dashboard should contain profiles with key metrics for the role. Below should be tables and graphs with key data available for the role, visualization of the park with units and notifications from System.

Personalization

- The dashboard must be customizable by the administrator based on user roles:
    - **Main Manager:**
        - profiles - key performance indicators (KPIs),
        - All reports,
        - notifications about delays in processing applications and contracts, approval of customers with not good credit history.
    - **Rent manager:**
        - profiles - applications, contracts, terminations, speed of closing applications and contracts,
        - tables and graphs - managing applications and contracts, tracking unit statuses, managing customers,

        - notifications about new applications, terminations, notifications about all letters received from customers, notifications of all deviations in the lease process and termination.
    - **Park manager:**
        - profiles - new damages and repairs, delay in repair deadline, cost of repairs, unpaid repairs.
        - tables and graphs - managing park and units, monitoring repairs and damages, dynamics of damages, managing maintenance tickets, tracking task completion
        - notifications about new damages, damages in the settlement process.
    - **Park Worker:**
        - profiles - new damages and repairs, delay in repair deadline,
        - tables and graphs - managing maintenance tasks and tracking ongoing repairs and damages,
        - notifications about new tickets, tickets not completed on time, new damages.
    - **Accountant:**
        - profiles - amount of money received, amount of accounts receivable, number of debtor customers
        - tables and graphs - financial analytics, payments, and outstanding debts, customer, amount of expected payments.
        - notifications about new invoices,
    - **Office Worker:**
        - tasks

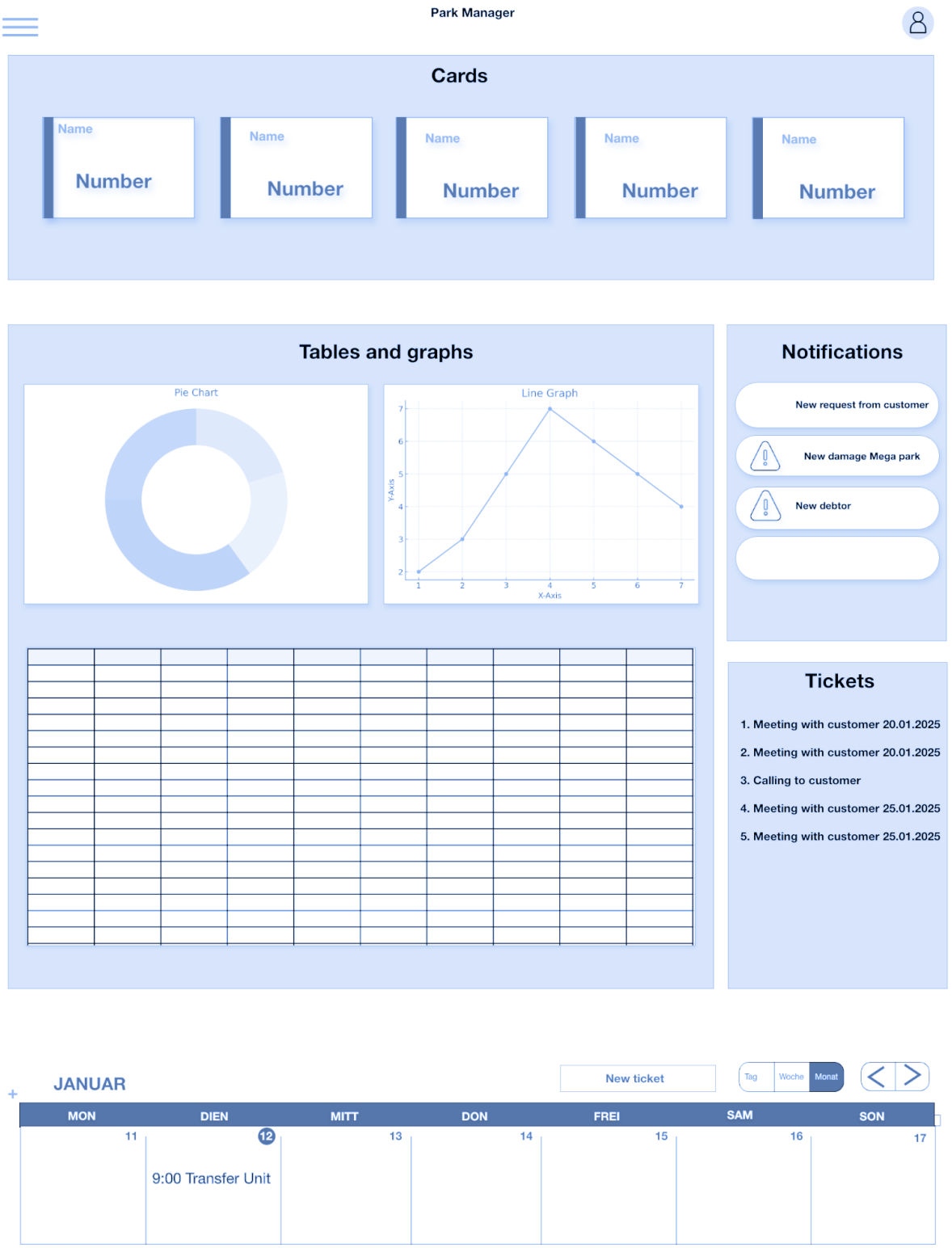
Key Features

- **Information profiles:** Displayed at the top of the dashboard.
    - Summarize key metrics for each user role.
    - Clicking a profile redirects the user to the corresponding menu section.
- **Tables and Graphs:**
    - Display detailed information relevant to the user's role;
    - Filters (parks, time, unique filters for table);
    - Search;
    - Drill-down functionality.
- **Notifications:**
    - A separate block on the dashboard, where all active notifications are listed.

- Each role has its own set of notifications;
- Notification is clickable and leads to the corresponding section of System.

User settings

Dashboard scheme



Dashboard, example:

https://drive.google.com/drive/folders/1CiyM3M9JESVuh9HpfUVDbDZn8bXQ-Pv_?usp=sharing

Contract:

https://drive.google.com/file/d/1tdRRMPE9-HzEnIp6HHrutQq432_WPyDt/view?usp=sharing

Financial:

https://drive.google.com/file/d/1BTVnU-dysCkW4A5OlVJSdUoepNmUnHyT/view?usp=sharing

https://drive.google.com/drive/folders/1CxHw3AgfNoR3iMWdisioANJ-F_XE6jSr

Customer:

https://drive.google.com/drive/folders/1VQdkmLbZo5EZF-nIsBQ5hC0EGFPWxPgE?usp=sharing

Unit:

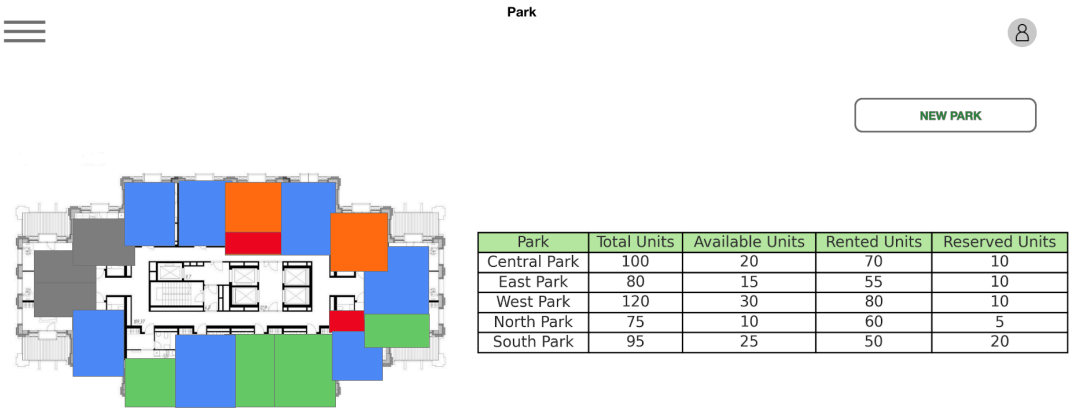
https://drive.google.com/drive/folders/1F8z7q6K3d6M5BzMNaal0BN5phwBghjaB?usp=sharing

Park:

https://drive.google.com/drive/folders/1QLuBsdQu9a7grpn0ntyZIh9FuO3swDPC?usp=sharing

Task:

https://drive.google.com/drive/folders/1pOl40mrsFLQ7cNjYxl6gajZ2dQUhbhDU?usp=sharing



Park	Total Units	Available Units	Rented Units	Reserved Units
Central Park	100	20	70	10
East Park	80	15	55	10
West Park	120	30	80	10
North Park	75	10	60	5
South Park	95	25	50	20

Park	Line	Unit Number	Previous Month Reading (kWh)	Current Month Reading (kWh)	Consumption (kWh)
South Park	Line A	U101	111	894	783
Central Park	Line C	U102	323	688	365
Central Park	Line A	U103	202	953	751
East Park	Line A	U104	402	737	335
South Park	Line B	U105	370	739	369
West Park	Line A	U106	459	869	410
Central Park	Line A	U107	438	691	253
East Park	Line C	U108	310	741	431
North Park	Line A	U109	286	816	530
East Park	Line A	U110	363	815	452
West Park	Line C	U111	274	532	258
North Park	Line A	U112	351	752	401
Central Park	Line B	U113	410	621	211
North Park	Line C	U114	205	549	344
West Park	Line B	U115	397	946	549
South Park	Line C	U116	346	986	640
Central Park	Line A	U117	152	672	520
South Park	Line B	U118	436	895	459
West Park	Line C	U119	144	890	746
Central Park	Line B	U120	119	955	836

Supplement with interface prototypes.

The tables have filters and a search string.

Template list

Color Scheme

(Should the color scheme match the corporate colors or is there a separate brand book for System?)

- Primary interface color: *Color name (#RRGGBB)*.
- Success operations color: *Color name (#RRGGBB)*.

- Error color: *Color name (#RRGGBB)*.
- Text block background: White (#FFFFFF)
- Text and background contrast ratio: At least 4.5:1.

We will later decide which colours we will use

Software Interfaces:

[Description of interaction with other systems, APIs, integrations. This will follow later on.]

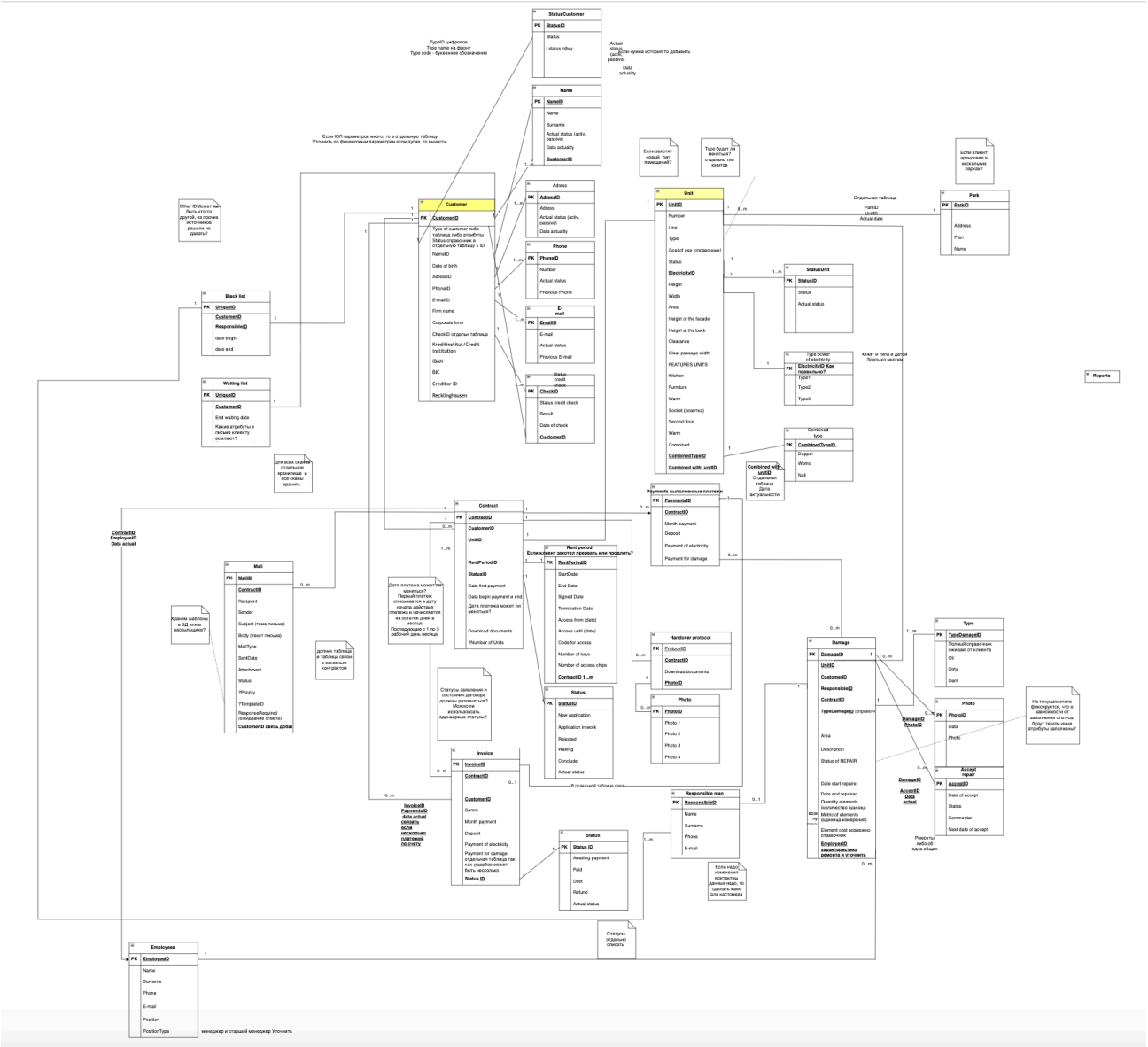
8. Data Requirements

Data Entities:

[Basic data entities, their attributes and data types]

Data Relationships:

(For developers) The diagram is available at the link:
https://drive.google.com/file/d/1yRPt5og8_KinJBecHMkzMRYraFTTuZ7n/view?usp=sharing



Data Flow Diagrams:

[Data flow diagrams that describe how data moves through the system]

Data Validation Rules:

[Data validation rules, such as input restrictions or format checks]

9. Workflow and Process Diagrams

Use Case Diagrams:

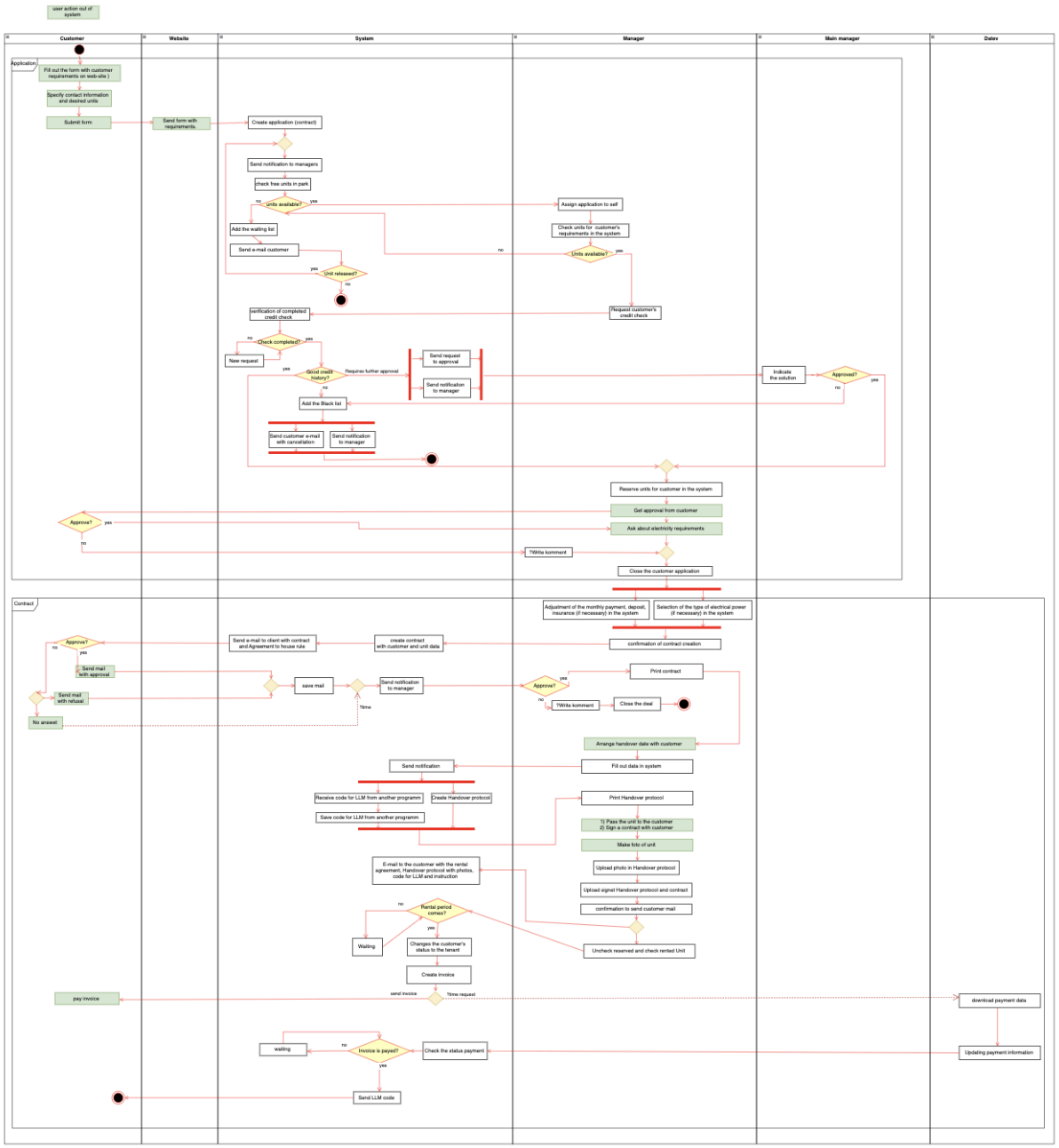
[Diagrams describing user interaction with the system]

Process Flow Diagrams:

Main process - Rent

The diagram is available at the link:

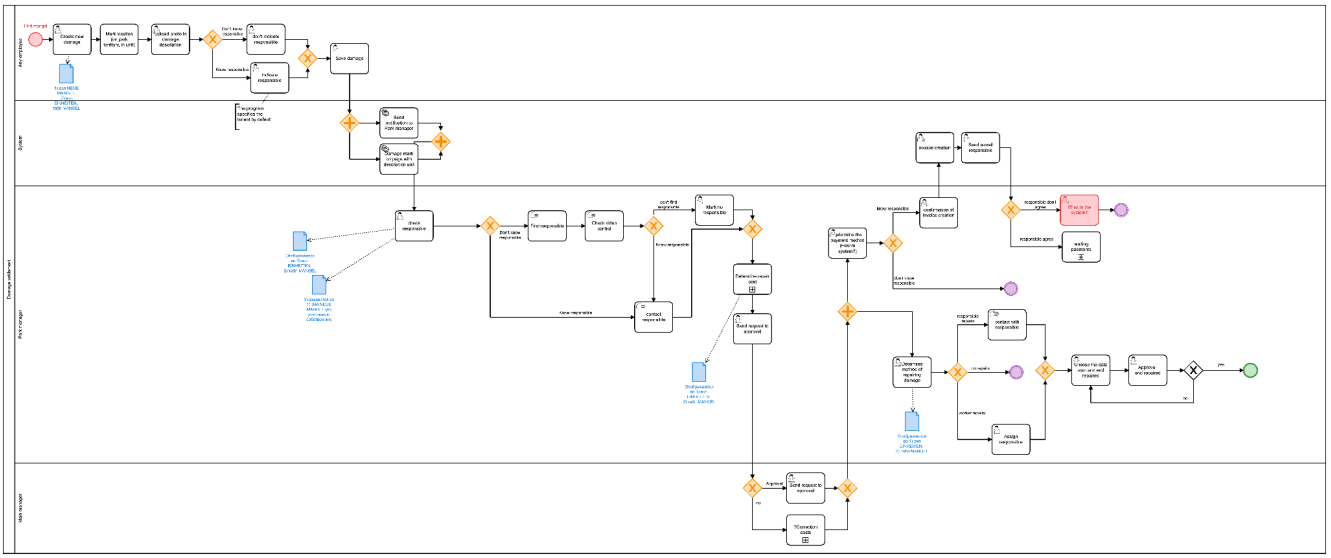
https://drive.google.com/file/d/132F-PO5drONwVSTSDQh8DaFrCo2OnsAx/view?usp=sharing



Damage control

The diagram is available at the link:

https://drive.google.com/file/d/1ccydY7gRpL2qThz1NvUWElqRwhi5QhUb/view?usp=sharing

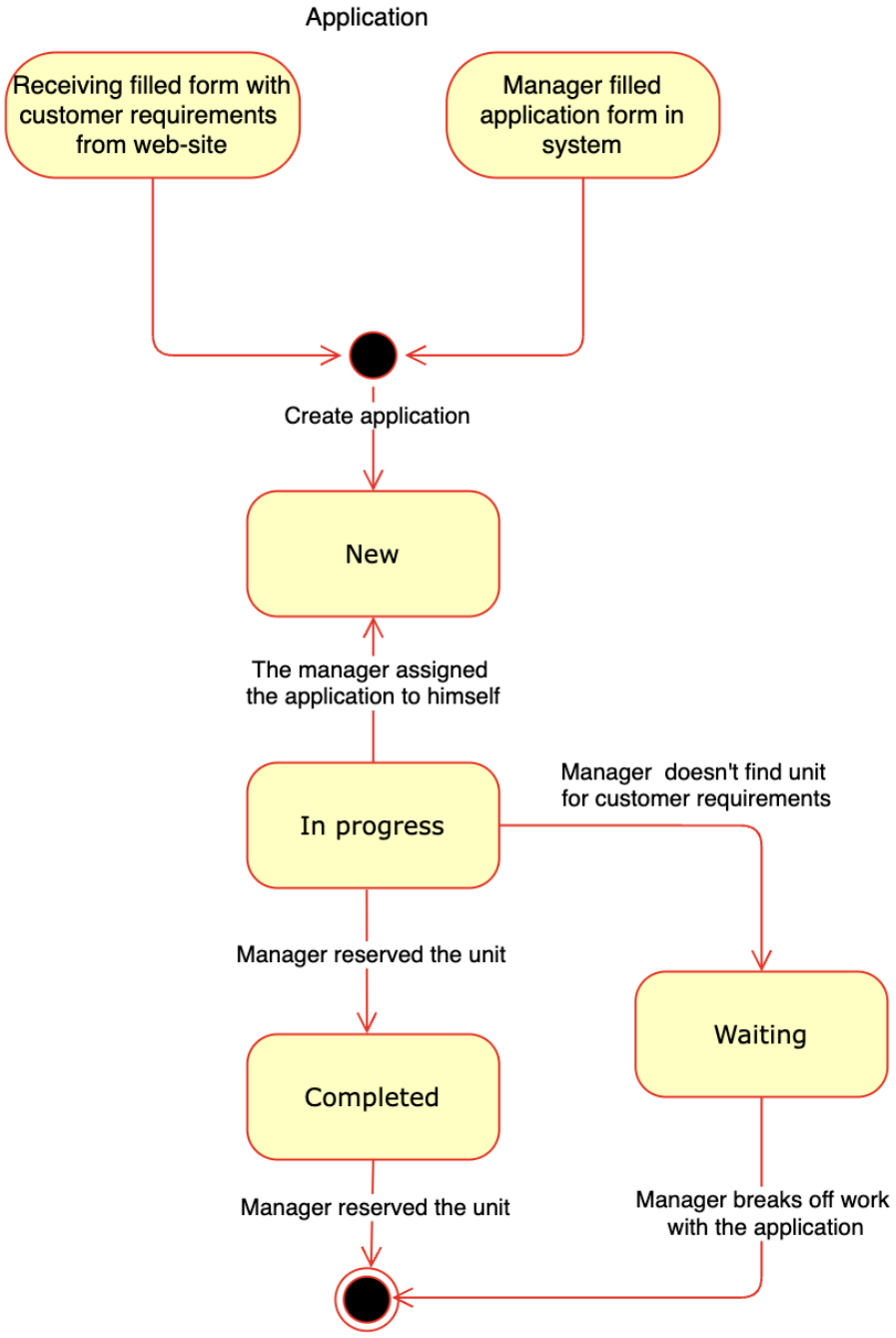


State Diagrams:

Application

The diagram is available at the link:

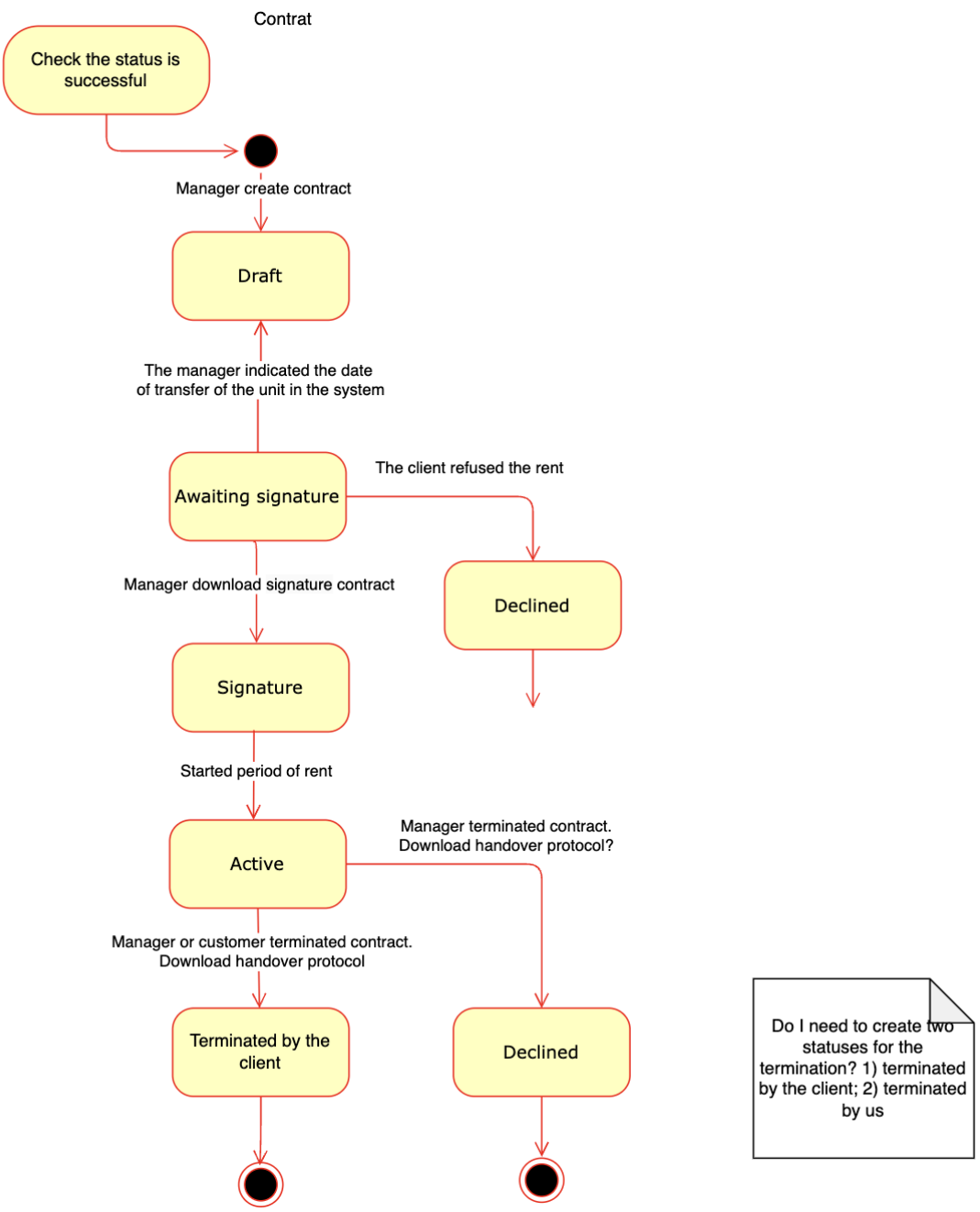
https://drive.google.com/file/d/1ujcicVREr7X-ce4TCiaUvPoHuRaS3aCy/view?usp=sharing



Contract

The diagram is available at the link:

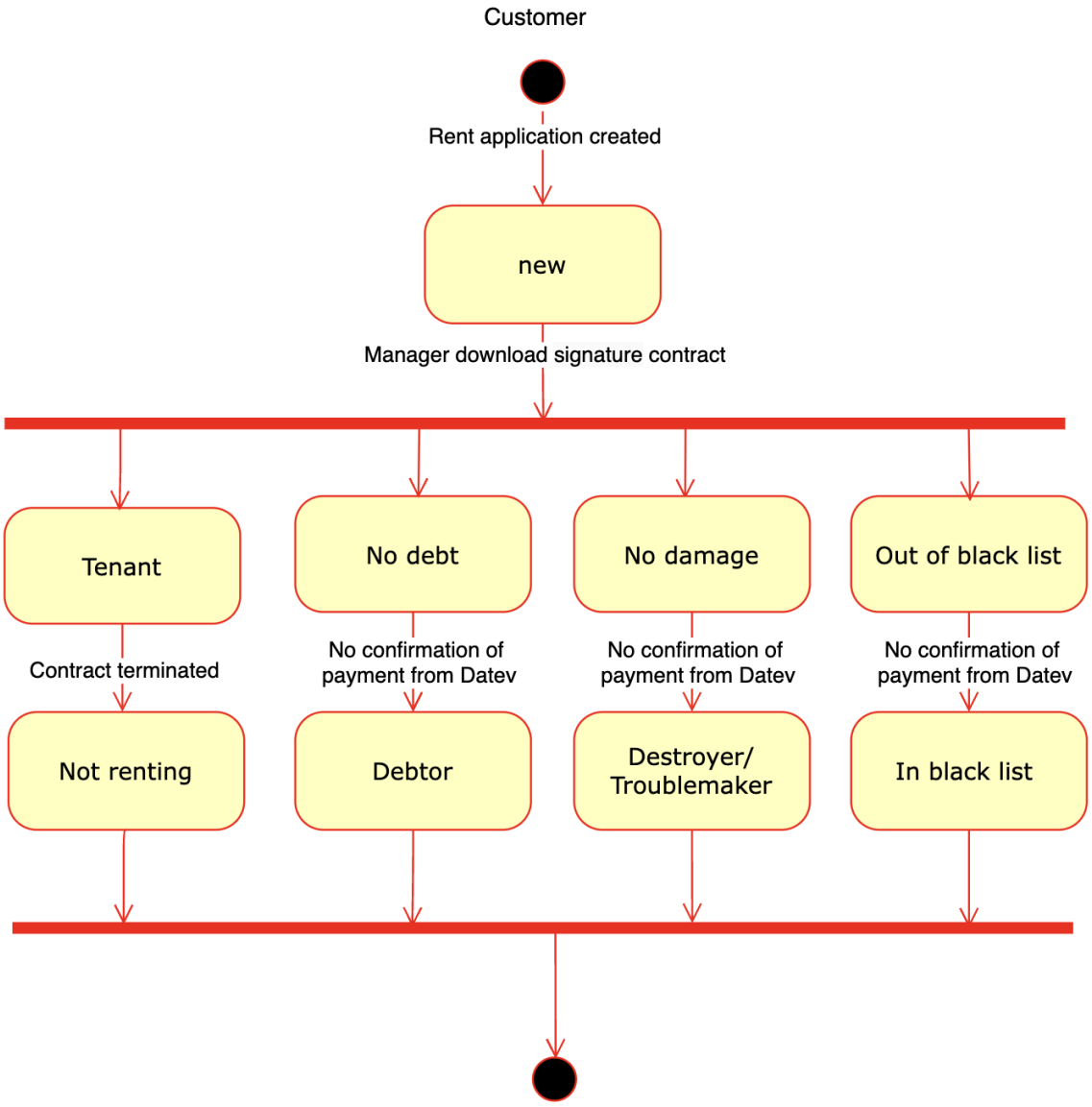
https://drive.google.com/file/d/1Pppb0SsdOyOGVIF3A_UMvfK98zULZior/view?usp=sharing



Customer

The diagram is available at the link:

https://drive.google.com/file/d/1cTnZ-OTFZl0OPiapiuF4eE07u_gro7El/view?usp=sharing



10. Reports

The reports module is designed to provide actionable insights into the financial, operational, and customer-related aspects of property rental and management. It enables stakeholders to monitor performance, track key metrics, and identify issues such as overdue payments or prolonged vacancies. Users can generate predefined or custom reports with filters by location, date, property type, or customer category. The module supports real-time and scheduled reporting, offering export options in PDF, Excel, and CSV formats. Role-based access ensures that each user views only the reports relevant to their responsibilities, maintaining security and data integrity.

The Report templates is available at the link:

https://docs.google.com/spreadsheets/d/16wKdTNXEs7lYDEnHkBEd-c_pK2eMW4Ts/edit?usp=sharing&ouid=110888704172854537327&rtpof=true&sd=true

11. Reference Data

1. General Requirements for Reference Data

1.1. Data Storage

- All reference data must be stored in the database with the ability to update without modifying the system code.
- Data must support versioning (tracking of changes over time).

1.2. Access and Management

- Access to edit reference data must be restricted by roles (e.g., only administrators can modify reference data).
- The system must support export/import functionality (e.g., CSV, JSON, Excel).
- Local reference books must inherit values from the master reference book but allow modifications where necessary.

1.3. Access and Management

1.3. Relationship Between Reference Data and System Objects

- The system should support a two-level directory system that provides standardization of property types and prices at a global level, as well as the ability to customize locally at the property park level.
- The master reference book should store base information (e.g., property types and their standard attributes).
- Local reference books (specific to each unit) should contain customized parameters relevant to individual units.
- Reference books must be linked to other system entities (e.g., property statuses, tenants, payments).

1.4. Change Audit

- All changes to reference data must be logged (who made the change, when, and what was modified).
- Change history should be accessible to administrators.

Use cases in section 5 Script 23.

12. Role model

1. Key Principles of the Role Model

Flexibility in Configuration

- The system must support both predefined standard roles and the ability to create custom roles.
- The administrator must have the ability to edit permissions for existing roles.
- It should be possible to assign individual permissions to users, regardless of their role.

The Role Model Should Be Based on RBAC (Role-Based Access Control)

- Access to system functionality is determined by role groups.
- A user can belong to one or multiple roles.
- Each role defines a set of allowed actions.

Granularity of Access Rights

- The settings must support access levels to system sections, actions, and objects.
- Ability to group permissions (e.g., "View", "Edit", "Delete", "Administer").

Independent Permission Configuration

- It should be possible to assign additional individual permissions to users on top of the role model.

2. Standard Roles and Basic Permissions

2.1. List of Predefined Roles

Role	Description	Basic Permissions
Admin	System administrator	Full access to all functions, user management, role and permission settings
Main Manager	Head manager	Park management, rental control, financial data review
Rental Manager	Rental manager	Tenant management, contract signing, rental status updates
Park Worker	Park employee	Repair and maintenance tasks, handling requests

Account (Accountant)	Accountant	Payment management, financial operations, integration with accounting systems
Office Worker	Office worker	Handling documents, customer requests, customer support

2.2. Examples of System Permissions

Permission	Description
View Tenants	Allows viewing the list of tenants
Edit Tenant Data	Allows editing tenant information
Create Contracts	Access to rental contract creation
View Payments	Access to payment history review
Edit Payments	Ability to modify payment information
Manage Tasks	Allows creating, editing, and closing tasks for employees
Configure Tariffs	Access to rental and utility tariff modifications
System Administration	Access to user management, reference data, and role settings

3. Functional Requirements for Role and Permission Management

3.1. Role Management

Creating, Editing, and Deleting Roles via the Interface

- The administrator must be able to create new roles through the system.
- A set of permissions can be assigned to each role.
- Permissions for existing roles can be edited.

Assigning Roles to Users

- The ability to assign one or multiple roles to a user.
- Ability to change a role without deleting the user.

3.2. Individual Permission Configuration

Additional Permissions for Specific Users

- The administrator must have the ability to grant special permissions to a user, even if they do not have the corresponding role.

- Example: An accountant can be given access to tenants, even though it is not part of their standard role.

Changing Permissions Without Modifying Roles

- If it is necessary to extend access for a specific user, additional permissions can be assigned without modifying their role.

3.3. Object-Level Access Control

Access by Parks (Locations)

- The ability to configure user access only to specific parks or locations.
- Example: A rental manager can only work with tenants in one park, while the main manager can manage all parks.

Access to System Sections

- The ability to restrict access to specific modules (e.g., disable the "Finance" section for a park worker).

4. Business Rules for the Role Model

Principle of Least Privilege (PoLP)

- By default, new users must have the minimum level of access.
- Only administrators can assign additional roles.

Logging Changes

- All changes to roles and permissions must be recorded in the system (who, when, to whom access was assigned or removed).

Backward Compatibility

- Changes to permissions in the system should not disrupt user operations (e.g., removing a role should not block access to necessary data).

Permission Inheritance

- If a user receives multiple roles, their permissions should be combined (the least restrictive policy should apply).

5. Role and Permission Management Interface

Section "Roles and Permissions"

- Table with existing roles.
- Buttons: "Add Role", "Edit", "Delete".

Role Creation Form

- Fields: Role name, Description, Checkboxes for available permissions.
- Button: "Save".

Assigning Roles to Users

- User list.
- Button "Change Permissions", which opens a window with role settings.

Individual Access Configuration

- A form that allows manually selecting additional permissions for users.

Appendices

Open Issues and TBDs:

[List of questions and items that require clarification]

An assessment is needed to create an interactive map of the park with the ability to view each unit and go to the unit page without leaving the park.
Or use a ready-made similar solution.